

**LAPORAN PTAKTIKUM BASIS DATA
MEMBUAT DATABASE LSP MENGGUNAKAN MARIA DB**



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BAB I

PENDAHULUAN

Latar Belakang

Database merupakan kumpulan data yang tersusun secara sistematis sehingga mudah disimpan, dikelola, dan diakses kembali. Dalam pengelolaan data yang cukup banyak diperlukan sebuah sistem database agar proses penyimpanan maupun pencarian data menjadi lebih efektif. Salah satu perangkat lunak yang dapat digunakan adalah MariaDB.

Pada praktikum ini dilakukan pembuatan database `db_lsp` yang terdiri dari beberapa tabel yang saling berhubungan menggunakan Foreign Key. Selain membuat tabel dan mengisi data, praktikum juga mencakup pembuatan View dan penggunaan Subquery untuk menampilkan informasi tertentu dari database.

1 Membuat Database

Masuk ke MariaDB kemudian membuat database dengan nama *db_lsp*.

```
Microsoft Windows [Version 10.0.26200.6899]
(c) Microsoft Corporation. All rights reserved.

C:\Users\devia>cd C:\xampp\mysql\bin

C:\xampp\mysql\bin>mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 11
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE db_lsp;
Query OK, 1 row affected (0.002 sec)
```

2 Membuat Tabel

Membuat delapan tabel yaitu:

Asesor, Asesi, skema, unit_kompetensi, tuk, jadwal, pendaftaran, hasil_uji. Seluruh tabel dibuat menggunakan perintah CREATE TABLE dan dihubungkan menggunakan Foreign Key.

```
MariaDB [(none)]> USE db_lsp;
Database changed
MariaDB [db_lsp]> CREATE TABLE asesor(
  -> id_asesor INT AUTO_INCREMENT PRIMARY KEY,
  -> nama_asesor VARCHAR(100),
  -> no_hp VARCHAR(20),
  -> email VARCHAR(100)
  -> );
Query OK, 0 rows affected (0.026 sec)
```

```
MariaDB [db_lsp]> CREATE TABLE asesi(
  -> id_asesi INT AUTO_INCREMENT PRIMARY KEY,
  -> nama_asesi VARCHAR(100),
  -> jk ENUM('L','P'),
  -> alamat VARCHAR(200),
  -> no_hp VARCHAR(20)
  -> );
Query OK, 0 rows affected (0.017 sec)
```

```
MariaDB [db_lsp]> CREATE TABLE skema(  
  -> id_skema INT AUTO_INCREMENT PRIMARY KEY,  
  -> nama_skema VARCHAR(100),  
  -> jenis_skema VARCHAR(50)  
  -> );  
Query OK, 0 rows affected (0.044 sec)
```

```
MariaDB [db_lsp]> CREATE TABLE unit_kompetensi(  
  -> id_unit INT AUTO_INCREMENT PRIMARY KEY,  
  -> id_skema INT,  
  -> kode_unit VARCHAR(20),  
  -> nama_unit VARCHAR(100),  
  -> FOREIGN KEY(id_skema)  
  -> REFERENCES skema(id_skema)  
  -> );  
Query OK, 0 rows affected (0.045 sec)
```

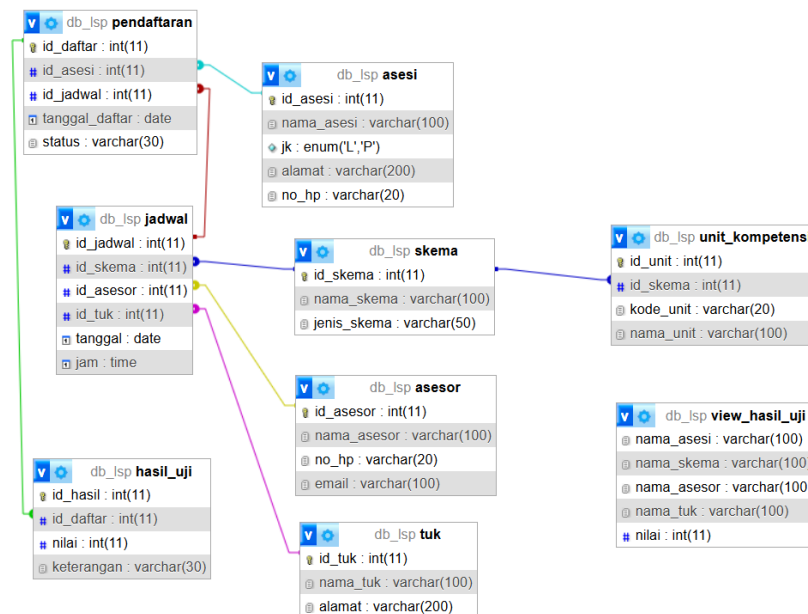
```
MariaDB [db_lsp]> CREATE TABLE tuk(  
  -> id_tuk INT AUTO_INCREMENT PRIMARY KEY,  
  -> nama_tuk VARCHAR(100),  
  -> alamat VARCHAR(200)  
  -> );  
Query OK, 0 rows affected (0.023 sec)  
  
MariaDB [db_lsp]> CREATE TABLE jadwal(  
  -> id_jadwal INT AUTO_INCREMENT PRIMARY KEY,  
  -> id_skema INT,  
  -> id_asesor INT,  
  -> id_tuk INT,  
  -> tanggal DATE,  
  -> jam TIME,  
  -> FOREIGN KEY(id_skema)  
  -> REFERENCES skema(id_skema),  
  -> FOREIGN KEY(id_asesor)  
  -> REFERENCES asesor(id_asesor),  
  -> FOREIGN KEY(id_tuk)  
  -> REFERENCES tuk(id_tuk)  
  -> );  
Query OK, 0 rows affected (0.055 sec)
```

```
MariaDB [db_lsp]> CREATE TABLE pendaftaran(  
  -> id_daftar INT AUTO_INCREMENT PRIMARY KEY,  
  -> id_asesi INT,  
  -> id_jadwal INT,  
  -> tanggal_daftar DATE,  
  -> status VARCHAR(30),  
  -> FOREIGN KEY(id_asesi)  
  -> REFERENCES asesesi(id_asesi),  
  -> FOREIGN KEY(id_jadwal)  
  -> REFERENCES jadwal(id_jadwal)  
  -> );  
Query OK, 0 rows affected (0.029 sec)
```

```
MariaDB [db_lsp]> CREATE TABLE hasil_uji(  
  -> id_hasil INT AUTO_INCREMENT PRIMARY KEY,  
  -> id_daftar INT,  
  -> nilai INT,  
  -> keterangan VARCHAR(30),  
  -> FOREIGN KEY(id_daftar)  
  -> REFERENCES pendaftaran(id_daftar)  
  -> );  
Query OK, 0 rows affected (0.025 sec)
```

3. Menginput Data

Menginput data ke setiap tabel sesuai soal, yaitu



```
MariaDB [db_lsp]> INSERT INTO asesor(nama_asesor,no_hp,email)
-> VALUES
-> ('Andi','0811111111','andi@gmail.com'),
-> ('Budi','0811111112','budi@gmail.com'),
-> ('Citra','0811111113','citra@gmail.com'),
-> ('Dedi','0811111114','dedi@gmail.com'),
-> ('Eka','0811111115','eka@gmail.com');
Query OK, 5 rows affected (0.005 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
MariaDB [db_lsp]> INSERT INTO asesi(nama_asesi,jk,alamat,no_hp) VALUES
-> ('Ahmad Fauzi','L','Bandung','0811111111'),
-> ('Siti Nurhaliza','P','Bandung','0811111112'),
-> ('Rizky Pratama','L','Bandung','0811111113'),
-> ('Dinda Aulia','P','Bandung','0811111114'),
-> ('Fajar Nugraha','L','Bandung','0811111115'),
-> ('Putri Maharani','P','Bandung','0811111116'),
-> ('M. Iqbal','L','Bandung','0811111117'),
-> ('Nabila Putri','P','Bandung','0811111118'),
-> ('Yoga Saputra','L','Bandung','0811111119'),
-> ('Ayu Lestari','P','Bandung','0811111120'),
-> ('Reza Maulana','L','Bandung','0811111121'),
-> ('Intan Permata','P','Bandung','0811111122'),
-> ('Dimas Ramadhan','L','Bandung','0811111123'),
-> ('Nur Aini','P','Bandung','0811111124'),
-> ('Rian Kurniawan','L','Bandung','0811111125'),
-> ('Vina Oktaviani','P','Bandung','0811111126'),
-> ('Agus Setiawan','L','Bandung','0811111127'),
-> ('Lina Marlina','P','Bandung','0811111128'),
-> ('Bayu Prakoso','L','Bandung','0811111129'),
-> ('Devi Aliya','P','Bandung','0811111130');
Query OK, 20 rows affected (0.062 sec)
Records: 20 Duplicates: 0 Warnings: 0
```

```
MariaDB [db_lsp]> INSERT INTO skema(nama_skema,jenis_skema)
-> VALUES
-> ('Junior Web Developer','IT'),
-> ('Network Administrator','IT'),
-> ('Database Programmer','IT'),
-> ('Digital Marketing','Bisnis'),
-> ('Graphic Designer','Desain');
Query OK, 5 rows affected (0.006 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
MariaDB [db_lsp]> INSERT INTO unit_kompetensi(id_skema,kode_unit,nama_unit)
-> VALUES
-> (1,'UK81','HTML'),
-> (1,'UK82','CSS'),
-> (1,'UK83','JavaScript'),
-> (1,'UK84','PHP'),
-> (2,'UK85','TCP/IP'),
-> (2,'UK86','Routing'),
-> (2,'UK87','Switching'),
-> (2,'UK88','Firewall'),
-> (3,'UK89','SQL'),
-> (3,'UK10','ERD'),
-> (3,'UK11','Normalisation'),
-> (3,'UK12','Stored Procedure'),
-> (4,'UK13','SEO'),
-> (4,'UK14','Google Ads'),
-> (4,'UK15','Social Media'),
-> (4,'UK16','Content Marketing'),
-> (5,'UK17','Photoshop'),
-> (5,'UK18','Illustrator'),
-> (5,'UK19','CorelDraw'),
-> (5,'UK20','UI Design');
Query OK, 20 rows affected (0.055 sec)
Records: 20 Duplicates: 0 Warnings: 0
```

```
MariaDB [db_lsp]> INSERT INTO tuk(nama_tuk,alamat)
-> VALUES
-> ('TUK Bandung','Bandung'),
-> ('TUK Jakarta','Jakarta'),
-> ('TUK Bogor','Bogor'),
-> ('TUK Bekasi','Bekasi'),
-> ('TUK Depok','Depok');
Query OK, 5 rows affected (0.026 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
MariaDB [db_lsp]> INSERT INTO jadwal(id_skema,id_asesor,id_tuk,tanggal,jam)
-> VALUES
-> (1,1,1,'2026-08-01','08:00:00'),
-> (2,2,2,'2026-08-02','09:00:00'),
-> (3,3,3,'2026-08-03','10:00:00'),
-> (4,4,4,'2026-08-04','08:30:00'),
-> (5,5,5,'2026-08-05','09:30:00'),
-> (1,2,1,'2026-08-06','08:00:00'),
-> (2,3,2,'2026-08-07','09:00:00'),
-> (3,4,3,'2026-08-08','10:00:00'),
-> (4,5,4,'2026-08-09','08:30:00'),
-> (5,1,5,'2026-08-10','09:30:00');
Query OK, 10 rows affected (0.048 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
MariaDB [db_lsp]> INSERT INTO pendaftaran(id_asesi,id_jadwal,tanggal_daftar,status)
-> VALUES
-> (1,1,'2026-07-20','Lulus'),
-> (2,2,'2026-07-20','Lulus'),
-> (3,3,'2026-07-20','Lulus'),
-> (4,4,'2026-07-20','Lulus'),
-> (5,5,'2026-07-20','Lulus'),
-> (6,6,'2026-07-21','Lulus'),
-> (7,7,'2026-07-21','Lulus'),
-> (8,8,'2026-07-21','Lulus'),
-> (9,9,'2026-07-21','Lulus'),
-> (10,10,'2026-07-21','Lulus'),
-> (11,1,'2026-07-22','Lulus'),
-> (12,2,'2026-07-22','Lulus'),
-> (13,3,'2026-07-22','Lulus'),
-> (14,4,'2026-07-22','Lulus'),
-> (15,5,'2026-07-22','Lulus'),
-> (16,6,'2026-07-23','Lulus'),
-> (17,7,'2026-07-23','Lulus'),
-> (18,8,'2026-07-23','Lulus'),
-> (19,9,'2026-07-23','Lulus'),
-> (20,10,'2026-07-23','Lulus');
Query OK, 20 rows affected (0.034 sec)
Records: 20 Duplicates: 0 Warnings: 0
```

```
MariaDB [db_lsp]> INSERT INTO hasil_uji(id_daftar,nilai,keterangan)
-> VALUES
-> (1,85,'Kompeten'),
-> (2,78,'Kompeten'),
-> (3,90,'Kompeten'),
-> (4,70,'Belum Kompeten'),
-> (5,88,'Kompeten'),
-> (6,92,'Kompeten'),
-> (7,75,'Belum Kompeten'),
-> (8,80,'Kompeten'),
-> (9,86,'Kompeten'),
-> (10,95,'Kompeten'),
-> (11,77,'Kompeten'),
-> (12,82,'Kompeten'),
-> (13,89,'Kompeten'),
-> (14,68,'Belum Kompeten'),
-> (15,91,'Kompeten'),
-> (16,84,'Kompeten'),
-> (17,73,'Belum Kompeten'),
-> (18,87,'Kompeten'),
-> (19,93,'Kompeten'),
-> (20,79,'Kompeten');
Query OK, 20 rows affected (0.075 sec)
Records: 20 Duplicates: 0 Warnings: 0
```

Menampilkan Data

Melakukan pengecekan data menggunakan perintah:

sql

SELECT * FROM nama_tabel;

```
MariaDB [db_lsp]> SELECT *FROM asesor;
+-----+-----+-----+-----+
| id_asesor | nama_asesor | no_hp | email |
+-----+-----+-----+-----+
| 1 | Andi | 0811111111 | andi@gmail.com |
| 2 | Budi | 0811111112 | budi@gmail.com |
| 3 | Citra | 0811111113 | citra@gmail.com |
| 4 | Dedi | 0811111114 | ded@gmail.com |
| 5 | Eka | 0811111115 | eka@gmail.com |
+-----+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [db_lsp]> SELECT *FROM asesi;
+-----+-----+-----+-----+-----+
| id_asesi | nama_asesi | jk | alamat | no_hp |
+-----+-----+-----+-----+-----+
| 1 | Ahmad Fauzi | L | Bandung | 0811111111 |
| 2 | Siti Nurhaliza | P | Bandung | 0811111112 |
| 3 | Rizky Pratama | L | Bandung | 0811111113 |
| 4 | Dinda Aulia | P | Bandung | 0811111114 |
| 5 | Fajar Nugraha | L | Bandung | 0811111115 |
| 6 | Putri Maharani | P | Bandung | 0811111116 |
| 7 | M. Iqbal | L | Bandung | 0811111117 |
| 8 | Nabila Putri | P | Bandung | 0811111118 |
| 9 | Yoga Saputra | L | Bandung | 0811111119 |
| 10 | Ayu Lestari | P | Bandung | 0811111120 |
| 11 | Reza Maulana | L | Bandung | 0811111121 |
| 12 | Intan Permata | P | Bandung | 0811111122 |
| 13 | Dimas Ramadhan | L | Bandung | 0811111123 |
| 14 | Nur Aini | P | Bandung | 0811111124 |
| 15 | Rian Kurniawan | L | Bandung | 0811111125 |
| 16 | Vina Oktaviani | P | Bandung | 0811111126 |
| 17 | Agus Setiawan | L | Bandung | 0811111127 |
| 18 | Lina Marlina | P | Bandung | 0811111128 |
| 19 | Bayu Prakoso | L | Bandung | 0811111129 |
| 20 | Devi Aliya | P | Bandung | 0811111130 |
+-----+-----+-----+-----+-----+
20 rows in set (0.001 sec)

MariaDB [db_lsp]> SELECT *FROM skema;
+-----+-----+-----+
| id_skema | nama_skema | jenis_skema |
+-----+-----+-----+
| 1 | Junior Web Developer | IT |
| 2 | Network Administrator | IT |
| 3 | Database Programmer | IT |
| 4 | Digital Marketing | Bisnis |
| 5 | Graphic Designer | Desain |
+-----+-----+-----+
5 rows in set (0.001 sec)
```